MIL-I-82356/26
4 April 1969

MILITARY SPECIFICATION

INSTRUMENTS, SIMULATED, AIRCRAFT TRAINING DEVICES;
GYRO ATTITUDE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The technical requirements for procuring the instrument described herein shall consist of this document and the issue in effect of Specification MIL-I-82356.

1. SCOPE

1.1 This specification covers the detail requirements for simulated aircraft attitude-gyro indicators for use in aircraft training devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-5133	Indicator, Attitude, Type J-8
MIL-I-6501	Indicator, Attitude, External Remote Vertical Gyro Stabilized, Type B-1A
MIL-I-7819	Indicator, Control, Gyro Horizon, Type MB-3 (P-1 Automatic Pilot)
MIL-I-7906	Indicator, Gyro Horizon, Manually Caged, Electric, Type MB-1
MIL-I-21203	Indicator, Attitude, Spherical, Direct Reading, Fast Erecting, Electric Gyro
MIL-I-21559	Indicator, Attitude, ARU-5/A

FSC 6930

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MIL-I-23043	Indicator, Attitude, Two Inch Self-Contained Standby
MIL-I-23084	Indicator, Attitude, Remote, 3-Inch Case
MIL-I-23085	Indicator, Attitude Remote 5-Inch Case
MIL-I-23524	Indicator, Attitude-Director, Remote
MIL-I-26359	Indicator, Attitude, Remote ARU-4/A
MIL-I-27193	Indicator, Attitude ARU-2B/A
MIL-I-27623	Indicator, Attitude, Remote Standby, 2 Inch
MIL-I-27680	Indicator, Attitude ARU-12/A Remote
MIL-I-27709	Indicator, Attitude ARU-14/A, Remote
MIL-I-27719	Indicator, Attitude ARU-13/A, Remote
MIL-I-82356	Instruments, Simulated, for Aircraft Training Devices, General Specification For

STANDARDS

Military

MS 17313	Indicator-Gyro Horizon, Electric Type MB-1
MS 17373	Indicator, Attitude, Two-Inch Self-Contained
MS 17400	Indicator, Attitude, Remote 3-Inch Case
MS 17401	Indicator, Attitude, Remote 5-Inch Case
MS 18028	Indicator, Attitude-Director, Remote
MS 25339	Indicator, Attitude; ARU-5/A
MS 27150	Indicator, Attitude, Remote, Standby

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MS 28073	Indicator, Control, Gyro Horizon Type MB-3 (P-1 Automatic Pilot)
MS 28086	Indicator, Attitude, Spherical, Direct Reading, Fast Erecting, Electric Gyro
MS 33545	Case-Instrument, 5 x 5 1/4, Standard Dimensions For
MS 33556	Case, Aircraft Instrument, 3 1/4 Inch Size, Standard Dimensions For

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

3. REQUIREMENTS

3.1 Operational instruments.- The requirements for the operational instruments being used as simulated instruments are as specified in the applicable operational military specifications designated in Table II.

3.2 Simulated instruments.- The requirements of MIL-I-82356 and this specification shall apply.

3.2.1 Design basis.- Each simulated and operational aircraft attitude-gyro indicator shall conform to the applicable operational instrument specified in Table I of this specification. The pitch indicating section shall be driven by a motor-generator through a 250 to 1 gear reduction which shall also drive a control transformer. The roll section shall be driven by a motor-generator through a 300 to 1 gear reduction which shall also drive a control transformer. The control transformers shall be electrically zeroed with zero degrees pitch and roll attitude indication.

3.2.2 Case.- With the exception of length, which shall not exceed the limits specified in the footnotes, cases shall be in accordance with Table I of this specification, as applicable.

3.2.3 Wiring.- Wiring shall be in accordance with the schematic referenced in Table I as applicable.

3.2.4 Pitch trim adjustment.- A hand operated knob identical in appearance and function to the hand operated knob on the operational instrument for adjusting the pitch trim indicator shall be provided for each simulated instrument, as applicable.

3.2.5 Demand erection device.- A pushbutton, identical in appearance and function to the pushbutton on the operational instrument for activating a fast erection system when the instrument is operating, shall be provided for each simulated instrument, as applicable.

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3.2.6 Caging device.- A caging device identical in appearance and function to the caging device on the operational instrument shall be provided for each simulated instrument, as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-I-82356 and as specified in 4.2 through 4.3.

4.2 Simulated instrument tests.- The following tests shall be performed on each instrument referenced in Table I in addition to tests specified in the test requirements of MIL-I-82356.

(a) Backlash of gears

(b) Forward and reverse operation

(c) Running smoothness of gears

(d) No jumping or hesitation of displays.

4.2.1 Slow running tests.-

(a) Pitch indicating system - The pitch indicating system of each simulated instrument shall be operated continuously through three cycles consisting of level indication to 80° climb to 80° dive back to level flight. Each cycle shall be completed in approximately 60 seconds

(b) Roll indicating system - The roll indicating system of each simulated instrument shall be operated continuously through 720° to the right followed by 720° to the left. Each 720° roll shall be completed in approximately 60 seconds. The indicating system shall track smoothly without jumping or any discontinuities.

4.2.2 Fast running tests.-

(a) Pitch indicating system - The test of paragraph 4.2.1(a) above shall be repeated with each cycle being completed in approximately 5 seconds. The indicating system shall track smoothly and shall not lag the test transmitter synchro by more than 10 degrees

(b) Roll indicating system - the test of paragraph 4.2.1(b) above shall be repeated with the 360° rolls being completed in approximately 1 second. The indicating system shall track smoothly and shall not lag the test transmitter synchro by more than 10 degrees.

4.3 Operational instrument tests.- The test procedure for instruments referenced in Table II shall be in accordance with the test procedure of the applicable operational specification listed in Table II. These tests shall be performed under the test requirements of MIL-I-82356.

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5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery requirements of MIL-I-82356 shall apply.

6. NOTES

6.1 The notes of MIL-I-82356 shall apply.

CUSTODIANS:
ARMY - AV
NAVY - TD
AIR FORCE - 70

USER ACTIVITIES:
ARMY - AV
NAVY - TD
AIR FORCE - 11, 70

PREPARING ACTIVITY:
NAVY - TD
Project Number 6930-0066

TABLE I - MOTOR-GENERATOR, CONTROL TRANSFORMER DATA

DESIG-NATION	OPERATIONAL INSTRUMENT BEING SIMULATED	NOMINAL SIZE	*CASE TYPE	SCHEMATIC WIRING PER MIL-I-82356 FIG. NO.
-1 S	MIL-I-5133***	3 inch	MIL-I-5133	12
-2 S	MIL-I-23043***	2 inch	MS 17373	12 **
-3 S	MIL-I-21203***	5 inch	MS 28086-1	12
-4 S	MIL-I-21203***	5 inch	MS 28086-2	12
-5 S	MIL-I-21203***	5 inch	MS 28086	12 **
-6S	MIL-I-21203***	5 inch	MS 28086	12 **
-7 S	MIL-I-7906***	3 inch	MS 17313	12
-8 S	MIL-I-21559***	3 inch	MS 25339	12 **
-9 S	MIL-I-7819***	3 inch	MS 28073	12

* Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 8 inches without prior approval from the procuring activity.

** Designation denotes an integrally lighted instrument.

*** An ON-OFF flag shall be provided which will appear identical to the operational instrument and shall be activated by 26V DC with a maximum current consumption of 1 AMP. The flag activation system shall not overheat under continuous duty.

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TABLE II - OPERATIONAL INSTRUMENTS USED IN NON-AIRBORNE TRAINERS

DESIG-NATION	OPERATIONAL INSTRUMENT BEING USED IN SIMULATED EQUIPMENT	NOMINAL SIZE	CASE TYPE	INTEGRALLY LIGHTED
-1	MIL-I-26359	3 inch	MIL-I-26359	Yes
-2	MIL-I-27193	5 inch	MS 33545	Yes
-3	MIL-I-27623	2 inch	MS 27150-1	Yes
-4	MIL-I-27623	2 inch	MS 27150-2	Yes
-5	MIL-I-27623	2 inch	MS 27150-3	Yes
-6	MIL-I-27623	2 inch	MS 27150-4	Yes
-7	MIL-I-23524	5 inch	MS 18028	Yes
-8	MIL-I-27680	5 inch	MS 33545	Yes
-9	MIL-I-6501	5 inch	MIL-I-6501	No
-10	MIL-I-27719	3 inch	MS 33556	Yes
-11	MIL-I-27709	5 inch	MS 33545	Yes
-12	MIL-I-23084	3 inch	MS 17400	Yes
-13	MIL-I-23085	5 inch	MS 17401	Yes

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